

Kinnaird Mills

The engineering study and the finance memo disagreed by more than the project was worth, and neither author had made an arithmetic error.

Kinnaird Mills ran a containerboard machine in central Alabama that bought 1.15 million MMBtu of natural gas and purchased steam a year. The proposal was a combined heat and power retrofit: a gas turbine and heat recovery steam generator that would displace essentially all of it. Installed cost, \$30 million. Useful life, fifteen years.

Aditi Ramaswamy's engineering firm had done the study and reported the saving the way engineering firms report savings — in physical units and current prices. "Displaced purchased energy of 1,150,000 MMBtu per annum, valued at the current delivered price of \$4.10 per MMBtu." A footnote on the following page observed that the firm's own long-term outlook had delivered gas escalating at 3.2 percent a year, and that the study had not applied it, "as escalation assumptions are the client's to make."

Gordon Rasch, the CFO, had made them. Or rather, he had not. His memo took the \$4.715 million of annual saving from the study, subtracted the \$780,000 service contract, ran the MACRS schedule, discounted the result at Kinnaird's 9.4 percent cost of capital, and got a negative number. He circulated it with one sentence: "Does not clear. Revisit if gas moves."

Beatriz Sandoval, two years into the treasury group, had been asked to check the model before it went in the file. She found the arithmetic clean. What she could not get past was the discount rate.

Kinnaird's 9.4 percent came off a beta, a risk-free rate of 4.1 percent, and an equity risk premium — all of them nominal, all of them quoted by markets that price in inflation. The Treasury inflation-indexed market implied around 2.4 percent general inflation over the horizon. The savings in Rasch's model were \$4.715 million in year 1 and \$4.715 million in year 15, which is a forecast that gas costs the same in 2041 as it does today.

"He has held the numerator in today's money and put tomorrow's money in the denominator," she said to her manager. "For fifteen years."

There were two more things in the file that she was less sure about.

The service contract was \$780,000 a year, and it was a contract — a fixed schedule of payments through year fifteen, signed with the turbine manufacturer, with no escalator in it. Whatever happened to inflation, Kinnaird would pay \$780,000 in year fifteen.

And the depreciation deductions were what the Internal Revenue Code said they were. Seven-year MACRS on \$30 million produced a fixed schedule of dollar amounts. Nothing in that schedule would move if inflation came in at four percent instead of two.

EXHIBIT 1 — THE RETROFIT

Installed cost, year 0	\$30.0m
Useful life	15 years
Displaced purchased energy	1,150,000 MMBtu/yr
Current delivered price	\$4.10/MMBtu
Expected energy price escalation	3.2%/yr
Service contract, fixed by contract	\$780,000/yr, years 1–15
Expected general inflation	2.4%/yr
Cost of capital, nominal	9.4%
Tax rate	25%
Depreciation	7-year MACRS on the full \$30m

The turbine is assumed to have no salvage value at the end of year 15.

EXHIBIT 2 — SEVEN-YEAR MACRS ON \$30 MILLION (\$ MILLIONS)

YEAR	1	2	3	4	5	6	7	8
Percentage	14.29	24.49	17.49	12.49	8.93	8.92	8.93	4.46
Deduction	4.287	7.347	5.247	3.747	2.679	2.676	2.679	1.338

Years 9 through 15 carry no depreciation.

ASSIGNMENT

- 1 Reproduce Rasch's number. Confirm that his arithmetic is right and say precisely which assumption is doing the damage.

- 2 Build the project in nominal terms: escalate what escalates, hold fixed what is fixed, and discount at 9.4 percent. Report the NPV.
- 3 Now build the same project in real terms. Derive the real discount rate, state what happens to each of the three cash flow streams when you convert them, and report the NPV. Your two answers must agree to the dollar. If they do not, find the stream you treated inconsistently.
- 4 Two of the three streams in this project are fixed in nominal dollars and one is not. Say which is which and why, and explain what a real-terms model has to do to the service contract and to the MACRS deductions.
- 5 Rasch's memo says "revisit if gas moves." Given your answer, has gas already moved for the purposes of this decision, and what would you put in front of him?